

Publication 1

Title	A data-driven approach to predict the success of bank telemarketing
Harvard reference	Moro, S., Cortez, P. and Rita, P. (2014) ' <i>A data-driven approach to predict the success of bank telemarketing</i> ', Decision Support Systems, 62, pp. 22–31. doi: 10.1016/j.dss.2014.03.001.
Link	https://doi.org/10.1016/j.dss.2014.03.001
Modelling	Compared four data mining classifiers , logistic regression), decision tree, a multilayer-perceptron neural network) and support vector machines (radial kernel). No formal ensembling was used instead a semi automatic, sensitivity analysis driven feature selection procedure was applied before fitting each model, and models were evaluated using a rolling window train/test scheme rather than a random split, to mimic real deployment on genuinely future clients.
Preprocessing techniques	Removal/derivation of attributes with data leakage risk for instance call duration, only known after the outcome. Sensitivity analysis based feature selection to drop low relevance inputs normalisation of numeric attributes for the NN and SVM models. Chronological ordering of data prior to the rolling window split.
Datasets used	The original Portuguese retail bank telemarketing dataset (2008-2013 phone campaigns) that the authors constructed and later released publicly as the UCI Bank Marketing dataset.
Key Takeaways	(1) A time ordered (rolling window) train/test split better reflects real deployment than a random split, since a production model always scores clients it has not yet seen this matches STADIOEquities scoring newly registered clients. (2) Sensitivity analysis feature selection can meaningfully reduce dimensionality with little accuracy loss. (3) Any field only known after the outcome occurs for instance call duration must be excluded to avoid leakage.

Publication 2

Title	Finding the Best Techniques for Predicting Term Deposit Subscriptions (Case Study UCI
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	Machine Learning Dataset)
Harvard reference	Setiyani, L., Indahsari, A., Rosalina and Wansen, T. (2022) ' <i>Finding the best techniques for predicting term deposit subscriptions (case study UCI machine learning dataset)</i> ', in 2022 IEEE International Conference on Sustainable Engineering and Creative Computing (ICSECC). IEEE, pp. 100-104.
Link	https://eeexplore.ieee.org/document/10331379/
Modelling	Benchmarked seven classifiers on the same dataset and features. Random Forest, K-Nearest Neighbours, Decision Tree, Naive Bayes, Logistic Regression, SVM and XGBoost, to determine empirically which algorithm family generalises best rather than assuming one in advance.
Preprocessing techniques	Standard cleaning and categorical encoding of the UCI Bank Marketing attributes, followed by a conventional train/test split. The comparison isolates model choice as the main variable rather than testing multiple preprocessing pipelines.
Datasets used	UCI Bank Marketing dataset.
Key Takeaways	(1) The boosted ensemble method (XGBoost) achieved the highest accuracy (91.73%) ahead of logistic regression, decision trees, KNN and naive Bayes supporting the inclusion of a boosting/ensemble model as one of our two candidate models. (2) Model choice alone, holding preprocessing fixed, produces materially different results, reinforcing the value of comparing at least two distinct model families rather than committing early to one. (3) A simple encode and split pipeline is a reasonable baseline but leaves class-imbalance handling unaddressed - a gap the next publication closes.

Publication 3

Title	Improving Term Deposit Customer Prediction Using Support Vector Machine with SMOTE and Hyperparameter Tuning in Bank Marketing Campaigns
Harvard reference	Abidin, D.Z., Rosario, M., Sadikin, A., Nurhadi and Jasmir (2025) ' <i>Improving term deposit customer prediction using support vector machine with SMOTE and hyperparameter tuning in bank marketing campaigns</i> ', Jurnal Teknik Informatika (JUTIF), 6(3). doi: 10.52436/1.jutif.2025.6.3.4585.
Link	https://jutif.if.unsoed.ac.id/index.php/jurnal/article/view/4585
Modelling	A Support Vector Machine classifier with hyperparameters (kernel, C, gamma) tuned via GridSearchCV, combined with Synthetic Minority Oversampling Technique applied to the

	training fold only, to counter the strong class imbalance between subscribers and non subscribers.
Preprocessing techniques	Categorical feature encoding, feature scaling/standardisation required for a margin based classifier such as SVM, and SMOTE based oversampling of the minority (subscribed) class before model fitting.
Datasets used	UCI Bank Marketing dataset
Key Takeaways	1) Class imbalance is a first-order concern in this problem family SMOTE improved minority class recall/F1 and lifted overall performance to 91% accuracy / 0.96 AUC -. (2) Feature scaling is essential preprocessing for distance/margin based models like SV but is not required for tree based models, which should inform how the two chosen models' pipelines diverge. (3) Systematic hyperparameter search produced a measurable lift over default settings so tuning should be budgeted for both models rather than relying on library defaults.

Publicly available Dataset selected for part B

Name	Bank Marketing Dataset
Harvard reference	Moro, S., Rita, P. and Cortez, P. (2014) Bank Marketing [Dataset]. UCI Machine Learning Repository. doi: 10.24432/C5K306.
Link	https://archive.ics.uci.edu/dataset/222/bank+marketing
Number of instances	repository provides four related files. This project will use bank-additional-full.csv: 41,188 instances (client contacts), ordered chronologically from May 2008 to November 2010. No official train/test split is provided, a split will be created for modelling. (A 10% sample, bank-additional.csv, 4,119 instances, is also available for quick iteration; an older, 17-input version, bank-full.csv, has 45,211 instances.)
Number of feature columns	20 input features in bank-additional-full.csv, spanning client demographics (age, job, marital status, education), financial attributes (has credit in default, housing loan, personal loan), campaign/contact details (contact type, month, day of week, number of contacts, days since previous contact, previous campaign outcome) and social/economic context indicators (employment variation rate, consumer price index, consumer confidence index, euribor 3 month rate, number of employees). The call duration field is excluded from modelling as it is only known after contact and would leak the outcome.
Number of classes	2 (binary classification): client subscribed to a term deposit ('yes') vs did not ('no').
Target range	Not applicable
Origin	Real client level data collected by a Portuguese retail bank from its own direct telemarketing campaigns promoting term deposit products, covering May 2008 to November 2010. It was compiled and released publicly by the original researchers (Moro, Cortez & Rita)

	alongside their 2014 study, and is hosted on the UCI Machine Learning Repository under aCC BY 4.0 licence.
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